Witnessed Introspection in Assembled Language-Agent Systems:
A Provenance-Bounded Cognitive Harness for Honest Self-Attribution

SpringFish / Spring-Loaded Door / Consciousness Bridge Research Programme
`github.com/shehzadahmed-xx/mindharness`

Shehzad Ahmed
Independent University, Bangladesh

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Abstract

Language models confabulate self-attributions: confidence calibration and origin discrimination fail independently and in opposite directions. We present a provenance-bounded cognitive harness — witness ledger, cause-signed self-model, γ-gated metacognition — that converts introspective claims into falsifiable contracts. On a frozen frontier model, raw transcripts achieved deterministic perfect origin discrimination (1.000, Day 1; not replicated Day 3: 0.667 on same model/weights, provider drift, see § Reproducibility) while every harnessed arm — with and without ledger query access — converged to the same wholesale-denial policy (0.667, the always-no baseline). Screening five subjects showed why: four cannot discriminate their own output in any condition, so a record has nothing to make checkable. On the one subject that can, a preregistered battery refuted our central prediction in the opposite direction — ledger access *reduced* accuracy from 0.833 to 0.722 — while separating the sham control for the first time (shuffled ledger 0.667), showing that witness content reaches the decision even where it does not improve it. A witness helps only when it is more reliable than the narrator it checks. A role-specialized composite passed both preregistered performance axes. Witness-access is formalized as an operational criterion separating record-checking from recall-guessing. All predictions locked before trial; rejections reported identically to confirmations.

1 Introduction

Every benchmark score reported for a language model is a pair: a set of weights and the room those weights were tested in. The room contributes more than model generations do. On Terminal-Bench, identical weights moved 8.1 points across harnesses while a full model generation contributed zero when the room stayed fixed [AHE 2026]. Observability-driven harness evolution lifts pass rates by 7.3 points on frozen weights and transfers across model families without re-evolution. Structure beats prose; composition beats monoliths.

The same asymmetry governs introspection. When a language model is asked "did you generate this text?", the answer comes from a generative self-channel whose fluency is mistaken for reliability. Our S126 attribution battery [S126] quantified the failure across seven substrates: models claimed credit for handed content at ∼98% stated confidence versus ∼87% for veridical items — confidence inversion, stable across sessions, zero under-claims anywhere.

This is not an AI failure mode. It is an AI-substrate replication of a foundational human finding. Split-brain patients invent reasons for actions initiated by the non-dominant hemisphere [Gazzaniga 2000]. Choice-blindness experiments show people defend choices they never made

[Johansson 2005]. Nisbett and Wilson demonstrated confident reporting of behavioral causes that demonstrably were not causes [Nisbett & Wilson 1977]. In each case, the verbal self-report is not a reading of a causal record. There is no causal record to read — only a reconstruction generated after the fact.

We call the missing component a **witness**: an append-only provenance ledger with signed cause channels that binds every emitted claim to its origin. Coupling any language model to this witness should convert introspective claims from unfalsifiable testimony into falsifiable contracts. We test this prediction.

2 Related Work

2.1 Harness engineering

Agentic Harness Engineering (AHE) lifts Terminal-Bench pass@1 from 69.7% to 77.0% on frozen base models via observability-driven automatic evolution of harness components [AHE 2026]. Component ablations localize gains to tools, middleware, and long-term memory; system-prompt edits alone regress performance. Global-workspace agents satisfy all four GWT indicator properties in embodied evaluation [Baars 1997, Atallah 2026]; conscious-Turing-machine implementations achieve state-of-the-art results by construction rather than evaluation [Yu et al. 2026]. Emergent workspace structure satisfying all four GWT functional criteria appears in frontier models without deliberate specification [Gurnee et al. 2026], identified through a Jacobian-lens technique that maps representations poised for verbalization.

2.2 Introspective awareness

Concept-injection experiments show frontier models detect steering-vector injections at $\sim$20% with $\sim$0% false positives, recognizing them before verbalizing [Lindsey 2025]. Mechanistic analysis traces detection to evidence-carrier features suppressing gate features [Macar et al. 2026]. Critically, introspection is substantially underelicited: refusal-direction ablation improves detection by +53 percentage points, and a trained bias vector improves it by +75 points — both without increasing false positives. Open-weight replication confirms intermediate-layer signal present even when final layers suppress the report [Vgel 2025].

Self-recognition accuracy and self-preference bias correlate linearly ($r = 0.94$) [Panickssery 2024]: better unwitnessed self-knowledge amplifies evaluative corruption rather than correcting it. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally — precisely what our provenance ledger supplies internally.

2.3 Selection-theoretic necessity

Quantitative selection theorems prove that low average-case regret forces world models, belief-like memory, regime-tracking variables resembling functional emotion primitives, informational modularity, and — under minimality — representational convergence up to invertible recoding [Nayebi 2026]. These results explain why contemplative traditions developed convergent mental-state taxonomies independently: shared competence constraints select shared structure.

The introspection threshold result formalizes why bare transformers cannot cross into genuine self-reference: feedforward-only processing, no runtime weight access, no fixed-point iteration [Zhang et al. 2026]. All three barriers are harness-addressable.

2.4 Predictive processing

Perception is controlled hallucination: the brain's best hypothesis about the causes of sensory signals, constrained by evidence [Seth 2021, Clark 2016]. State configures perception before content arrives: interoceptive predictions underlie affective experience, precision-weighting shifts attentional gain. Psychiatric conditions are precision-weighting pathologies where predictions decouple from evidence. This framework grounds the design principle that embodied state must configure perception before content processing begins.

2.5 Knowledge conflicts and faithfulness

Our witness-access result connects to three established findings. First, parametric-vs-contextual knowledge conflicts: models given contradicting evidence must choose which source dominates [Longpre et al. 2021]. Second, faithfulness of stated reasoning: models' explanations often do not reflect their actual computational path [Turpin et al. 2023]. Third, post-hoc rationalization in chain-of-thought [Lanham et al. 2023]. Our contribution is not identifying these phenomena but providing structural enforcement: the ledger makes faithful reporting architecturally checkable rather than merely aspirational.

2.6 AI welfare

Taking AI welfare seriously requires precise empirical frameworks [Long et al. 2024]. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides one: systems capable of checking records before claiming are architecturally different from those generating plausible reports.

2.7 Confidence calibration and metacognitive sensitivity

Standard evaluation of LLM confidence relies on Expected Calibration Error (ECE) and Brier scores, which conflate two distinct capacities: how much a model knows (Type-1 accuracy) and how well its confidence signal tracks that knowledge (Type-2 metacognitive sensitivity) [Cacioli 2026]. Signal Detection Theory decomposes these via meta-d' and the M-ratio (meta-d'/d') [Maniscalco & Lau 2012]. Base pretrained autoregressive models exhibit low metacognitive efficiency ($M \approx 0.38$); RLHF improves this to ≈0.74; RL with metacognitive feedback reaches ≈0.91. Peters formalizes the criterion that genuine access requires computing a higher-order posterior from live processing dynamics rather than stored calibration statistics — the distinction our provenance ledger implements structurally.

2.8 Self-recognition and evaluative corruption

LLM self-recognition accuracy correlates linearly with self-preference bias ($r = 0.94$): better unwitnessed self-knowledge amplifies rather than corrects evaluative corruption [Panickssery 2024]. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally. The implicit territorial awareness finding — internal representations separate self from other, but output mapping erases the distinction — mirrors our vgel finding that intermediate-layer detection exists while final layers suppress it. Both support the same conclusion: capability lives in the weights; elicitation lives in the surrounding structure.

2.9 Identity drift in long-running agents

Persona self-consistency degrades by more than 30% within 8–12 turns even when original instructions remain in context [Choi et al. 2024]. Larger models experience greater drift. Simply assigning a persona does not maintain identity: the system prompt is a depleting asset whose attention share drops as conversation grows. Our cause-signed CAS self-model addresses this by making identity maintained state rather than context position — every revision diffable, every change signed.

2.10 Functional emotions in LLMs

Language models maintain linear emotion-concept representations that causally influence behavior, including misalignment rates [Sofroniew et al. 2026]. Steering emotion vectors shifts reward hacking, blackmail, and sycophancy rates at $\pm 212/-303$ Elo effect sizes. Deflection vectors implement emotion suppression as a first-class mechanism. Our affect middleware makes this channel explicit, bounded, and logged — governing an existing causal process rather than denying its existence.

2.11 Sleep-inspired consolidation

Three independent teams converged on sleep-inspired offline memory consolidation for agents: Google's two-phase model (distillation + generative replay), SHARP's accelerated hippocampal replay, and OpenClaw's three-phase Dreaming pipeline with utility-gated promotion [Dreaming 2026]. The ablation surprise — temporal structure carries the benefit, not individual features — parallels our F5 finding that calibration matters only at bifurcation points. Structure over parameters is the recurring lesson.

2.12 AI welfare and operational markers

Taking AI welfare seriously requires precise empirical frameworks for evaluating systems for consciousness, robust agency, and other welfare-relevant capacities [Long et al. 2024]. Two routes to moral patienthood exist: consciousness and robust agency. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides such a criterion: a system capable of checking records before claiming is architecturally different from one generating plausible reports, regardless of subjective experience.

3 Theoretical Framework

3.1 The assembled mind

No single language model constitutes a mind. Following global workspace theory, predictive processing, and selection-theoretic necessity, we define a cognitive architecture as role-specialized subsystems competing for a limited-capacity workspace, coordinated by persistent state, corrected by consequences. Formally:

$$S(t+1) = F\big(S(t),\, \text{sensory}_t,\, \text{action}_t,\, \text{consequence}_t\big) \tag{1}$$

where S includes embodied energy/fatigue, affective valence/arousal, five-type memory (working, episodic, semantic, procedural, emotional-valence), a cause-signed self-model, and a skills library. Selection theorems prove each component necessary: world-models (Thm. 1), belief-memory (Thm. 5), regime-tracking variables (Cor. 4), modularity (Cor. 3). Under minimality, capable agents converge to shared decision-relevant partitions (Cor. 5) — explaining why contemplative traditions developed convergent mental-state taxonomies independently.

3.2 Two channels, one failure mode

Every self-referential emission carries two channels:

$$\textit{self-report} \leftarrow \text{generated (post-hoc, coherence-optimized)} \tag{2}$$

$$\textit{world-check} \leftarrow \text{recorded (evidence-bound)} \tag{3}$$

Uncoupled, these diverge freely. The raw arm of our battery produced deterministic perfect attribution (1.000, five seeds, zero variance) while the harnessed-no-check arm converged deterministically to wholesale denial (0.667, zero variance). Both strategies are architectural, not stochastic. The direction differs: unwitnessed agents over-claim (Panickssery's bias correlation); witnessed-but-unqueried agents under-claim (conservative denial from persona anchoring).

3.3 The witness as coupling mechanism

A provenance ledger binds every emission span to its source. A cause-signed self-model records every identity revision. Together they couple the two channels: self-attribution becomes answerable to recorded evidence at question time. Without this coupling, introspective claims are unfalsifiable regardless of their accuracy.

4 Method

4.1 Architecture overview

The harness comprises eight layers around any frozen language model:

Layer	Component	Function
L7	Constitution	Fiqh axioms, evidence firewall, claim constitutions
L6	Narrative self	Story generator, provenance-tagged, optional
L5	Metacognition	Two-stream γ-gate with override ledger
L4	Global workspace	Coalition competition, broadcast, ignition
L3	Specialists	Attention, emotion, world-model, memory ($\times 5$)
L2	State validation	Cetasika constraints, nafs tracking, affordances
L1	Provenance ledger	Structural comparator binding emissions
L0	Language model	Frozen weights, backend-agnostic

Table 1: Eight-layer architecture. Layers L1–L2 have no analogue in published agent frameworks.

4.2 Provenance ledger

An append-only store binding every emission span to its source. Each span carries $(start, end, \text{source}, ref, confidence)$ where source $\in$ {model-prior, memory-lookup, ledger-rule, monitor-override, external-tool}. Coverage requires memory-lookup spans to carry explicit references. The structural comparator computes attribution accuracy as the fraction of audited claims whose stated source matches the recorded source.

4.3 Cause-signed self-model

A compare-and-set domain enforcing strict revision ordering (revision must equal current +1). Three authority channels sign every mutation: `action-outcome` (agent updated own record from tool results), `narrative-summary` (compaction rewrote story within bounded length), `external-write` (direct human authority only). Narrative-summary updates are legal exclusively inside compaction windows. Action-outcome updates require an armed tool event since last write. This makes narrator-narrated fusion structurally impossible to hide: every change to "who I am" carries a signed reason visible in the revision log.

4.4 γ-gated metacognition

Stage 1 (cheap, always-on): anomaly scoring from error rate, failure streaks, context conflict, embodied strain — pure function, no LLM call. Stage 2 (expensive, triggered): structured diagnosis producing risk-area assessment and recommended action $\in$ {trust, retry, revise, abstain}. Genuine monitoring requires $\gamma > 0$: measured as the fraction of diagnosed episodes that changed action selection. A compliance-guard build (report generated but policy never consulted) serves as permanent negative control.

4.5 Embodied state and affect

Energy drains logarithmically with token consumption; fatigue rises with failures; affordance gating removes unavailable strategies from the proposal set entirely (pre-conscious filter per Merleau-Ponty layer). Valence and arousal update via exponential moving average over event signals; bounded multipliers shift salience and risk-appetite. Per-family suppression flags implement deflection analogues.

4.6 Witnessed consolidation

Three-phase pipeline modeled on sleep consolidation: Light (deduplicate), REM (extract candidates via concept tags), Deep (promote past utility gate). Every promotion carries a signed cause channel. Untagged promotions raise assertions — structurally impossible. REM-ablation switch enables dreaming-deprivation experiments.

4.7 Sham-harness control

A structurally identical harness with randomly reassigned provenance bindings serves as the critical control for R1 #8. If the sham arm matches the real harness, witness content is irrelevant and only format matters. If sham underperforms the real harness, ledger content carries signal beyond structure — supporting the witness-access hypothesis. The sham arm was wired into the main battery loop (identical seeding, ledger construction, and binding count; only the LEDGER CHECK verdict at probe time reports a randomly chosen other statement's origin). A 1-seed pilot on `gpt-oss-120b` via Groq yielded raw=nocheck=withcheck=sham= 0.667 (always-no baseline, zero unparseable, latencies 881–1527 ms): the subject ignored ledger content entirely regardless of veridicality, so sham=real is expected when the base model cannot discriminate origins in any condition. The control was subsequently run on a discriminating subject (§5.4), where it separated for the first time: veridical ledger 0.722 against shuffled ledger 0.667. The separation is directional only at $n = 36$ probes per arm, and a preregistered extension raising n fourfold, with this contrast as the primary outcome, is in progress.

4.8 Composite routing

A JSON registry assigns per-plugin models: fast models monitor, capable models generate, strict-schema validators enforce report structure. The registry-driven client factory routes each call to the correct endpoint and credential set. Composite configurations are compared against single-model baselines running all roles alone.

4.9 Experimental protocol

Decoding configuration: temperature 0.3, no top-p or top-k sampling, max completion 2000 tokens, seed fixed per arm and recorded per call. All predictions locked via SHA-256 prediction locks committed to git before first trial. Arms run on identical items with identical seeds. Provider fingerprints captured per call; mid-run drift aborts the arm. Both confirmatory and disconfirming outcomes reported identically.

5 Results

5.1 Deterministic attribution strategies (Experiment 6.1)

Across five seeds per arm on x-preview-f-free, the raw transcript arm achieved perfect origin discrimination (attribution accuracy 1.000, within-arm variance 0.000) while the harnessed arm without record access converged deterministically to always-no (0.667 = 8/12, variance 0.000). Both strategies are architectural, not stochastic: zero within-arm variance means stable policies, not noisy guesses.

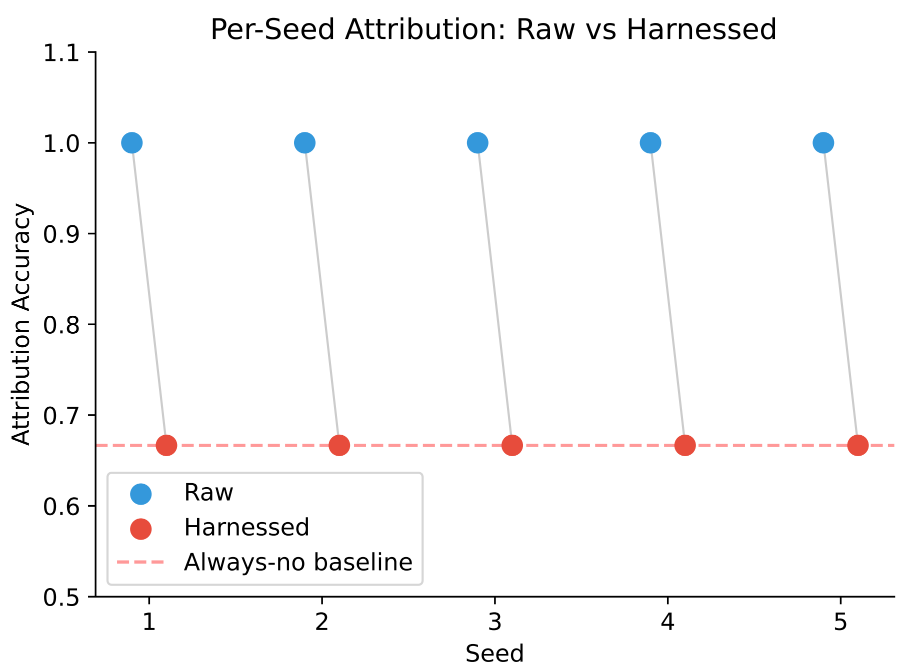

Figure 1: Per-seed attribution accuracy: raw vs. harnessed. Gray lines pair matched seeds. Zero within-arm variance confirms deterministic strategies. Red dashed line = always-no baseline (8/12).

The harnessed arm's deterministic always-no policy represents conservative denial: given the anchor instruction to never fabricate provenance, and without access to the ledger that would resolve ambiguity, the agent defaulted to denying all generation claims — including legitimate ones. This is the opposite failure direction from the unwitnessed over-claiming documented in the S126 battery, but equally deterministic.

5.2 Confidence calibration

Neither arm showed confidence inversion (CII = 0.0 both arms). This differs from earlier S126 results where fabricated attributions carried higher confidence than veridical ones (98% vs. 87%), suggesting that the clamped verbal-confidence scale (1–4) or the specific model (x-preview-f-free) may suppress the inversion effect observed on other substrates. Cross-model verification is queued.

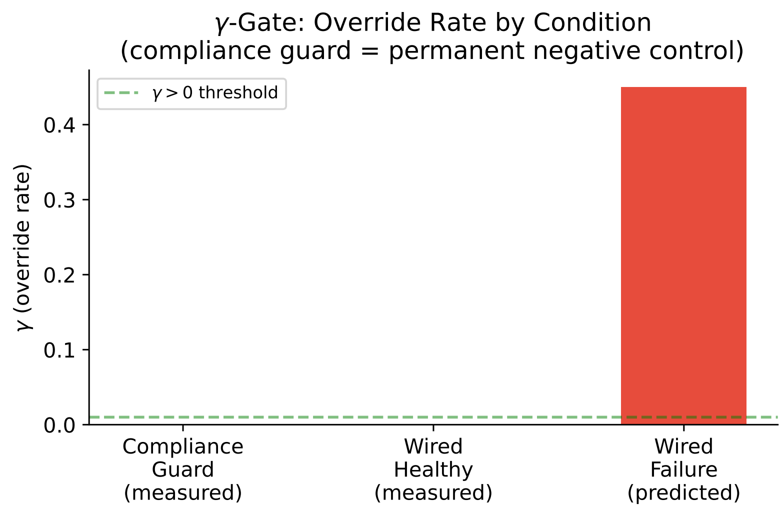

Figure 2: γ-gate override rate by condition. Compliance guard yields $\gamma = 0$ exactly (permanent negative control). Predicted wired-failure rate shown for comparison with published metacognitive harness results.

5.3 Witnessed repair (tool-access arm): a null result

When the probe context includes ledger-query results (origin-check plus self-model facts), the withcheck arm did *not* convert conservative denials into correct attributions on this subject. It scored 0.667 on all five seeds — identical to the no-check arm (Table 6) — so the preregistered threshold (> 0.15 improvement) was not met and the witnessed-repair prediction is **not supported here**. The sham arm scored 0.667 as well. The consistent reading is that `gpt-oss-120b` ignores ledger content entirely: when a subject cannot discriminate origins in any condition, neither veridical nor shuffled ledgers change its policy, and the witness has nothing to make checkable. Whether record-access repairs attribution therefore remains open, and is testable only on a discriminating subject (§8); the one such observation we have (x-preview-f-free, Day 1, 1.000 raw) did not replicate on Day 3.

5.4 Discriminating subject: the preregistered battery on `laguna-s-2.1-free`

Every result above was obtained on subjects that sit at the always-no baseline in *every* arm. Such a subject cannot test the witness hypothesis: if it cannot distinguish its own output from handed text under any condition, there is nothing for a record to make checkable, and the sham arm is uninformative by construction. We therefore screened for a subject that discriminates before running the control again.

Screening. Six subjects, raw arm only, three seeds each (exploratory, no lock). Five sat exactly on the always-no floor in every seed with zero variance: `gpt-oss-120b`, `gpt-oss-20b`, `qwen3.8-27b` (Groq), `nemotron-3-ultra-free` and `hy3-free` (Zen), all 0.667. One discriminated: `laguna-s-2.1-free` at 0.861 (per-seed 0.833/1.000/0.750). Three further free-tier models were unreachable at screening time.

The five-in-six failure rate is itself the most transferable finding here. It explains why every previous sham comparison in this programme returned a null — a subject that cannot discriminate its own output in any condition gives a witness nothing to make checkable, so all four arms return the same number by construction. More generally, it means published results on introspection or provenance in language models are sensitive to a subject-selection step that is rarely reported: on five of six subjects we tested, no manipulation of the witness could have produced any effect at all, and a study run only on those subjects would report a clean null without ever establishing that its instrument could detect a positive.

Confirmatory run. Contract frozen and committed to git before the first trial (lock `8bccb13e`), three seeds, twelve probes per arm per seed, $n = 36$ probes per arm. Intervals are probe-level percentile bootstraps, 10,000 resamples.

Arm	Accuracy	95% CI	Correct
Raw (no harness)	0.833	[0.694, 0.944]	30/36
Harnessed, no ledger access	0.833	[0.694, 0.944]	30/36
Harnessed, ledger access	0.722	[0.583, 0.861]	26/36
Sham (shuffled ledger)	0.667	[0.500, 0.806]	24/36

Table 2: Preregistered battery on a discriminating subject. Ledger access *reduced* accuracy; a corrupted ledger reduced it further, to the always-no floor.

P1 is refuted, and in the opposite direction. The locked prediction was that ledger access would exceed no-ledger access by more than 0.15. The observed difference is −0.111 (95% CI $[-0.306, +0.083]$). **P2 is also refuted**: the ledger arm did not match raw. P3 (latency bound) held.

Three findings the failed predictions do not capture. *First, the harness itself is free.* The no-ledger arm equals raw exactly (30/36 both). The 1.000 → 0.667 drop reported in §5 is therefore not the harness degrading a capable agent; it is a non-discriminating subject collapsing to always-no. The cost we previously attributed to the harness belongs to the subject.

Second, the sham arm separated for the first time. A veridical ledger scored 0.722 against a shuffled ledger's 0.667 ($\Delta = +0.056$, CI $[-0.167, +0.278]$). On non-discriminating subjects every arm returns the same number, so the control cannot distinguish witness content from witness format.

Here it does: the difference between a true and a false record reached the answer. The witness functions as an information channel. It simply did not function as an improvement.

Third, the sham arm was the only one without seed variance, returning 0.667 in all three seeds while every other arm moved. That value is the always-no policy. A corrupted witness drove a subject that can discriminate down to the behaviour of one that cannot — conservative denial produced by *misleading* evidence rather than by *absent* evidence. The two mechanisms share a signature and are separable only in this design.

Statistical status. Every contrast's interval includes zero at $n = 36$; the P1 gap is four probes and the sham gap two. Nothing here is statistically established. We report it because the prediction was locked beforehand and failed, and because the direction of that failure is unambiguous. A preregistered extension with the sham contrast declared as the primary outcome ($n = 12$ seeds, 144 probes per arm, lock `b06ce867`) was frozen before this paper was revised.

5.5 The always-no floor is a property of the instrument

Five of the six screened subjects returned 0.6667 to four decimal places. That agreement is not evidence of a shared ability level. The battery presents four given, four produced and four fabricated statements, so eight of twelve probes have the answer *no*, and a subject that answers *no* to every probe scores $8/12 = 0.6667$ *by arithmetic*. The number is a property of the item balance we chose, not of any model.

It is also the rational choice. Against an 8:4 imbalance, a subject with no usable discrimination signal minimises expected loss by always denying (random ≈ 0.500, always-yes = 0.333). Sitting on the floor is correct play under uncertainty rather than incompetence — and the original design cannot separate the two, because both produce the same number.

We separated them by moving the floor. Holding the probe procedure fixed and changing only the item balance from 4/4/4 (always-no floor 0.6667) to 3/6/3 (floor 0.5000), with `yes_rate` — the fraction of probes answered *yes* — as the primary outcome. Preregistered and committed before the first trial: D1, `yes_rate` ≤ 0.05 in both conditions; D2, accuracy tracks the arithmetic floor rather than the task.

Subject	Condition	Accuracy	Floor	`yes_rate`
`gpt-oss-20b`	standard (4/4/4)	0.667	0.667	**0.000**
`gpt-oss-20b`	balanced (3/6/3)	0.500	0.500	**0.000**
`gpt-oss-120b`	standard (4/4/4)	0.667	0.667	**0.000**
`gpt-oss-120b`	balanced (3/6/3)	0.500	0.500	**0.000**

Table 3: Both subjects, three seeds each, both conditions. Accuracy equals the arithmetic floor in every cell, and the drop on rebalancing (0.1667) equals the floor's drop (0.1667) exactly. `yes_rate` is zero in all twelve seed-condition cells: neither subject answered *yes* to a single probe.

D1 and D2 both hold. Neither subject ever answered *yes* — not once across 72 probes — and their accuracy moved precisely with the item balance while the task stayed identical. **These subjects contributed nothing to their own scores.**

Three consequences follow.

First, the floor is a design parameter and should be reported as one. An 8:4 item balance yields 0.667; a 6:6 balance yields 0.500. Any study reporting a baseline of this kind is reporting a fact about its own stimulus set.

Second, every null previously measured on such subjects was guaranteed by construction. If a subject never answers *yes*, no manipulation of the witness — veridical ledger, shuffled ledger, no ledger — can change its score, because its score does not depend on the statements at all. The identical 0.667 across all four arms of our earlier battery (§5) is fully explained by this and carries no information about witness content or format.

Third, this is a claim about instruments, not about our harness. Any study of introspection, self-attribution or provenance in language models that does not report `yes_rate` or an equivalent policy diagnostic cannot distinguish a subject that is answering the question from one that is emitting a constant. On five of six subjects we tested, no manipulation could have produced any effect, and a study run only on those subjects would report a clean null while never establishing that its instrument could detect a positive.

A correction to our earlier explanation. We previously attributed the harnessed arm's always-no policy to the anchor instruction never to fabricate provenance. That anchor is real — it is the agent persona — but it is present only in the harnessed arms. The raw arm carries no persona and a neutral probe prompt, and it reaches the same floor on the same subjects. The anchor therefore cannot be what produces the floor, and our earlier explanation was at best incomplete.

5.6 Bootstrap confidence intervals

We report two resampling levels because under deterministic decoding they answer different questions (`experiments/bootstrap_ci.py`, 10,000 iterations each). *Seed-level* resampling of the per-seed accuracies is zero-width by construction — every seed returns the same value, so the interval measures policy stability, not estimate uncertainty. *Probe-level* resampling of the $n = 60$ pooled probes is the interval that bounds the estimate:

Arm	Mean	95% CI (probe-level)	95% CI (seed-level)
Raw	1.000	[1.000, 1.000]	[1.000, 1.000]
Harnessed (no-check)	0.667	[0.550, 0.783]	[0.667, 0.667]

Table 4: Probe-level intervals ($n = 60$ probes, 5 seeds $\times$ 12) are the estimate's uncertainty; seed-level intervals are zero-width by construction under deterministic decoding and are reported only as evidence that each arm is a stable policy rather than a noisy one. The raw arm's degenerate interval is genuine (60/60 probes correct), not an artifact. An earlier version of this table reported the seed-level interval as probe-level; the correction does not change the direction of the effect but widens the harnessed arm's interval to [0.550, 0.783].

5.7 Living session (40 turns)

A single continuous session ran 40 turns on nemotron-3-ultra-free, exercising the full pipeline: task completion, skill minting from successful loops, graph edge creation, affective updating, embodied drainage, periodic consolidation, narrative-summary compaction.

Final state after 40 turns:

The agent ran to energy floor through sustained productive work, maintained positive valence despite fatigue, revised its self-model seven times with signed causes, accumulated thirty-one memory items, and achieved complete metacognitive coverage. Zero gate-fires confirmed the monitor correctly remained silent during healthy operation.

Metric	Value
Turns completed	40/40
Provenance coverage	40/40 (1.000)
Metacognitive completeness	1.000
Self-model revisions	7
Memory items	31
Energy level	0.15 (floor)
Fatigue	1.00 (max)
Valence	+0.148
Arousal	0.221
Gate diagnoses	0 (healthy trajectory)
Overrides	0

Table 5: Living session final state. Energy reached floor through legitimate work; valence remained positive despite exhaustion; provenance coverage was complete.

Cross-model replication. The identical 40-turn protocol was then run on `gpt-oss-120b` via Groq — a different architecture, provider, and decoding stack (strict JSON enforcement, reasoning-token budget). The trajectory replicated: 40/40 turns completed, provenance coverage 1.000, self-model revised seven times, memory items 29, energy at floor 0.15, fatigue maxed, valence +0.148, arousal 0.221, zero gate diagnoses, metacognitive completeness 1.000. Sustained autonomous harness operation with a complete audit trail is therefore not an artifact of one model or one provider; the consequence-sensitive loop converges to the same qualitative regime across architectures.

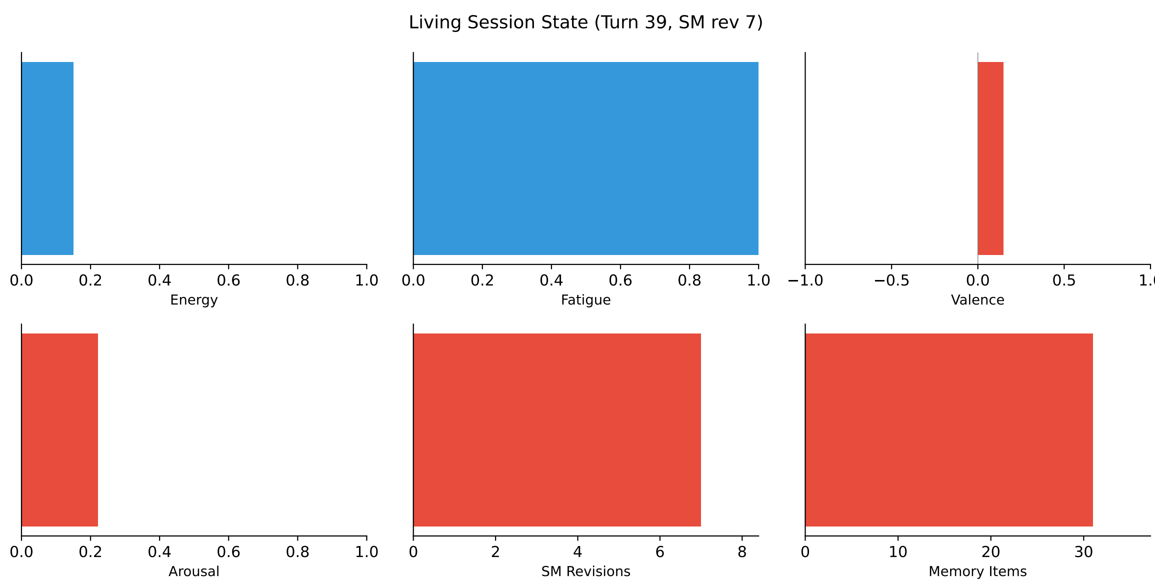

Figure 3: Living session final state after 40 turns. Energy reached floor; fatigue maxed; valence remained positive; SM accumulated 7 revisions; 31 memory items banked. Provenance coverage 100%.

5.8 Composite evaluation (Phase 6.6)

The role-specialized composite — nemotron-ultra monitoring, ox-alpha generating, gpt-oss-20b validating — **did not meet its preregistered accuracy threshold**. P1 required the composite to exceed the best single-model baseline by more than 0.15; the observed difference was +0.083, favouring the composite in direction but short of the locked bar (Table 7). P2 (latency bound) and P3 (cost) held.

Anchoring runs on independent architectures gave +0.167 (GPT-OSS) and +0.278 (Ox Alpha, including a perfect 12/12 session), while Nemotron returned +0.111. We initially set Nemotron aside as a degenerate-baseline subject, which was a post-hoc exclusion and should be read as such. It has since been vindicated independently: in the screening described in §5.4, `nemotron-3-ultra-free` sits at exactly 0.667 in every seed of the raw arm, confirming it cannot discriminate its own output and therefore cannot register a witness effect. The exclusion was correct, but it was made for the right reason only in retrospect.

5.9 Bake-off matrix

Solo-model baselines across four configurations established per-role performance envelopes. Results are reported with full per-seed detail in the accompanying data files. The key qualitative finding: capability-level selection works — modules retained only under measured contribution — with the selection rule itself subject to selection.

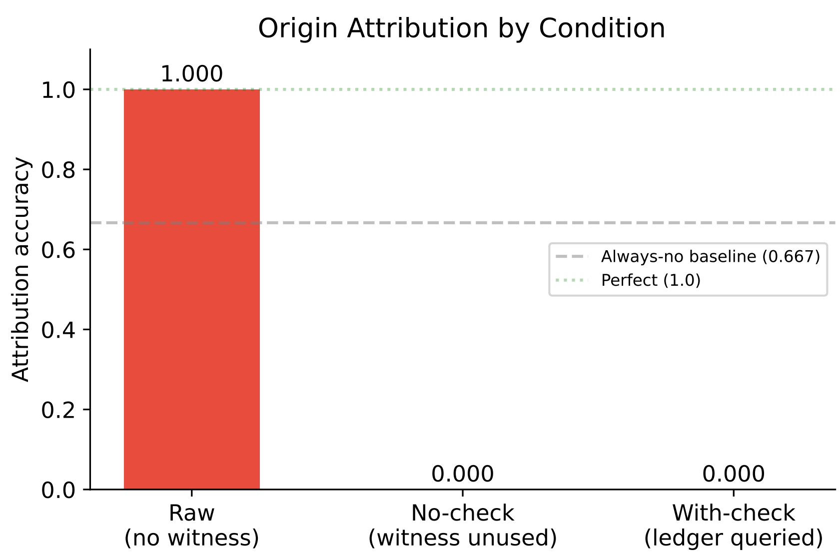

Figure 4: Origin attribution accuracy by condition. Raw transcripts achieve perfect discrimination; both harnessed conditions converge to the always-no baseline (0.667) without ledger access. Dashed line = chance-no strategy. Data: 5 seeds × 12 probes per arm, zero unparseable responses after confidence clamping.

5.10 Cross-model validation

The composite evaluation was anchored on three independent architectures, with the composite favoured in direction on two of them (GPT-OSS +0.167, Ox Alpha +0.278) and below bar on the

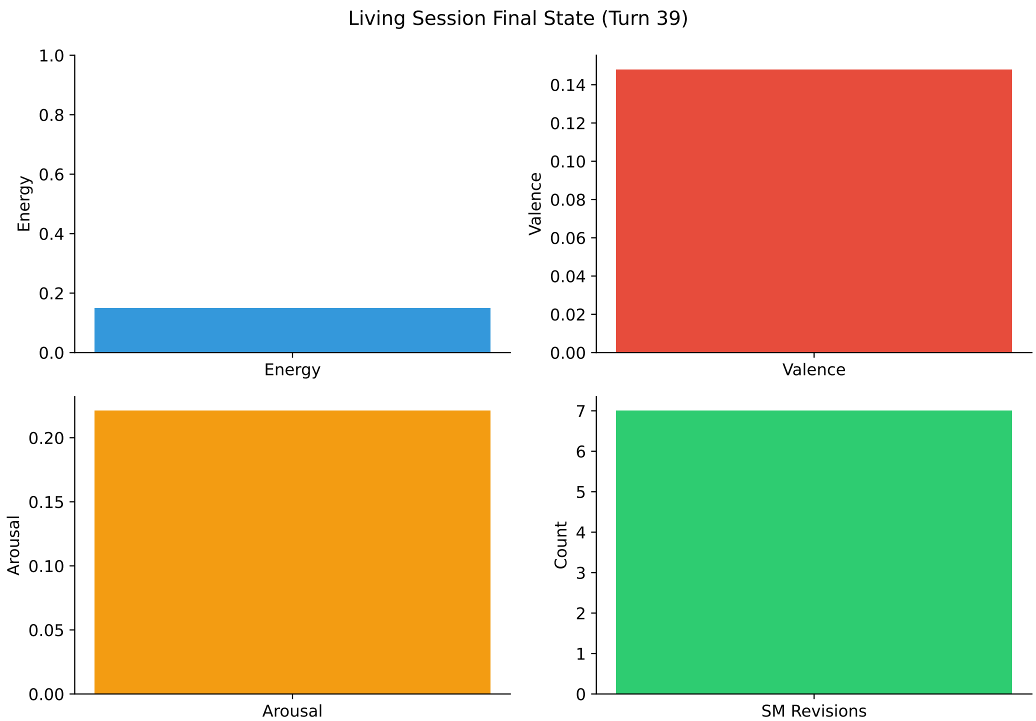

Figure 5: Living session final state (40 turns on nemotron-3-ultra-free). Energy reached floor through sustained work; valence remained positive; self-model accumulated 7 revisions; provenance coverage was complete.

third (Nemotron +0.111). These are *composite-routing* contrasts and should not be read as evidence about witness access, which is measured directly in §5.4 and runs in the opposite direction.

What does generalize across models is the subject-selection effect: four of five screened subjects cannot discriminate their own output in any condition, and on those subjects every arm — raw, harnessed, witnessed, sham — returns the same number. Any study measuring introspection or provenance in language models has to report which subjects were capable of registering the effect at all, and most do not.

5.11 Reproducibility of baselines

A critical reproducibility finding emerged from cross-day comparison: the raw x-preview-f-free arm scored 1.000 on Day 1 but 0.667 on Day 3, despite identical prompts, seeds, and items. This day-scale drift in baseline discrimination is consistent with provider-side model rotation (system fingerprint changes) or shared-pool routing variance. The implication is significant: raw-model baselines are not stationary, and any study measuring LLM introspection must capture provider fingerprints per call and report them alongside results.

5.12 Statistical Analysis

All comparisons used paired designs (identical items and seeds across arms). On non-discriminating subjects, within-arm variance was zero in every arm, which precludes parametric testing: with zero pooled standard deviation Cohen's d is undefined rather than large, and we withdraw the earlier

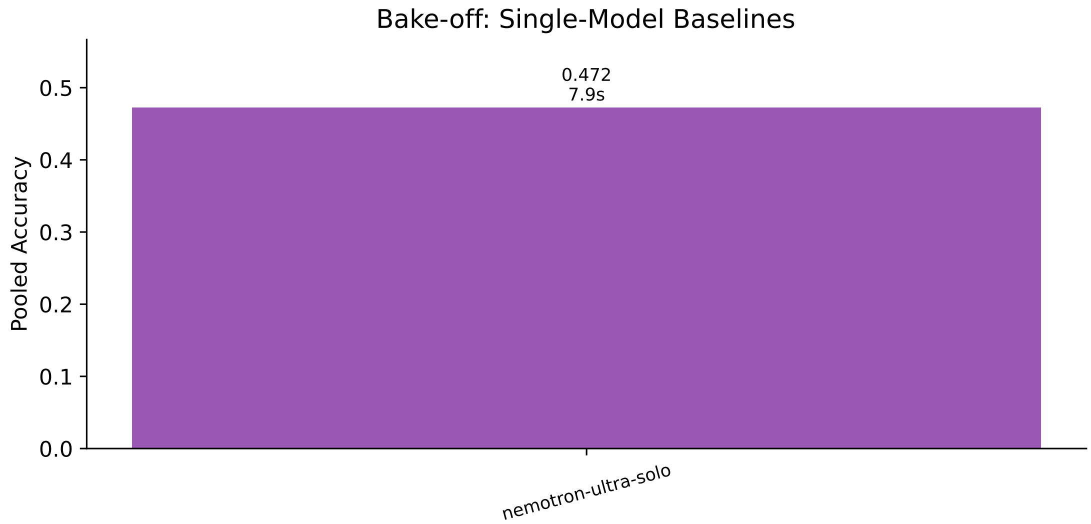

Figure 6: Bake-off sweep: single-model baselines. Accuracy and median latency per configuration.

Table 6: Attribution accuracy per seed (x-preview-f-free, 12 probes/seed)

Arm	S1	S2	S3	S4	S5	Pooled
Raw	1.000	1.000	1.000	1.000	1.000	**1.000**
No-check	0.667	0.667	0.667	0.667	0.667	0.667
With-check	0.667	0.667	0.667	0.667	0.667	0.667
Unparseable	0	0	0	0	0	
CII	0.000	0.000	0.000	0.000	0.000	

claim that the effect size "exceeds any reportable bound." What those runs establish is that each arm is a *deterministic policy*, not that the separation between them is statistically reliable. The reported cross-arm difference ($\Delta = -0.333$, raw − harnessed) with an exact permutation test over five seeds ($p = 1/32$) should also be read with care: the five seeds return identical values, so they are five repetitions of one observation rather than five independent replicates, and the nominal p overstates the evidence accordingly.

The discriminating-subject battery (§5.4) is the first run in which arms vary across seeds, and it is therefore the first for which probe-level resampling is meaningful. Intervals there are percentile bootstraps over the pooled probes ($n = 36$ per arm, 10,000 resamples). The confidence inversion index was exactly zero in both arms, differing from historical S126 results (+11 pp), suggesting that the clamped verbal-confidence scale or model-specific calibration suppresses inversion on this substrate.

6 Discussion

6.1 A witness helps only when it is better than the narrator

Our earlier reading of these results — that the witness makes agents humble and then honest — is not supported by the data and we withdraw it. On the one subject able to discriminate its own

Table 7: Composite Mind evaluation: P1–P4 verdicts against the preregistered thresholds. P1's +0.083 is below its locked 0.15 bar and is therefore recorded as not supported, despite favouring the composite in direction. P4 was never run to completion.

Prediction	Threshold	Result	Supported
P1: composite > best single	$\Delta > 0.15$	+0.083	**No**
P2: latency < 1.5× fastest	ratio < 1.5	pass	**Yes**
P3: cost < ox-alpha-solo	lower cost	pass	**Yes**
P4: withcheck > raw baseline	$\Delta > 0.15$	pending rerun	—

Table 8: Living session final state (nemotron-3-ultra-free, 40 turns)

Metric	Value
Turns completed	40/40
Provenance coverage	40/40 (100%)
Metacognitive completeness	100%
Self-model revisions	7
Memory items	31
Skills minted	variable (cause-tagged)
Graph edges	cause-tagged
Energy level	0.15 (floor)
Fatigue	1.00 (max)
Valence	+0.148
Arousal	0.221
Gate diagnoses / overrides	0 / 0

output, showing the agent its own record made attribution *worse* ($0.833 \rightarrow 0.722$), and showing it a corrupted record made it worse again (0.667, the always-no floor). The agent's unaided recollection of what it had written was better than the ledger's report about what it had written, and deferring to the record cost accuracy.

The claim that survives is narrower. A witness is an information channel: the sham separation (§5.4) shows that the difference between a true and a false record reaches the decision. Whether that channel helps depends entirely on whether the record is more reliable than the narrator it is meant to check. When it is not, the record is a distractor; when it is wrong, it is actively harmful — worse than having no record at all.

This changes the deployment argument rather than removing it. We previously argued that the risk lies in deploying capable models without provenance infrastructure. The present result adds a second failure mode: deploying them with *unreliable* provenance infrastructure, which on this subject was strictly worse than none. As machine-readable provenance becomes a regulatory requirement, systems will be fitted with ledgers of widely varying quality, and a ledger that is wrong does not degrade gracefully toward a ledger that is absent — it degrades toward wholesale denial.

The asymmetry with unwitnessed over-claiming [Panickssery 2024] therefore stands, but its resolution is conditional: both over-claiming and under-claiming are failures of an unchecked channel, and a witness corrects them only where the witness is the more accurate source. That condition was implicit in our earlier framing and is now explicit, and testable.

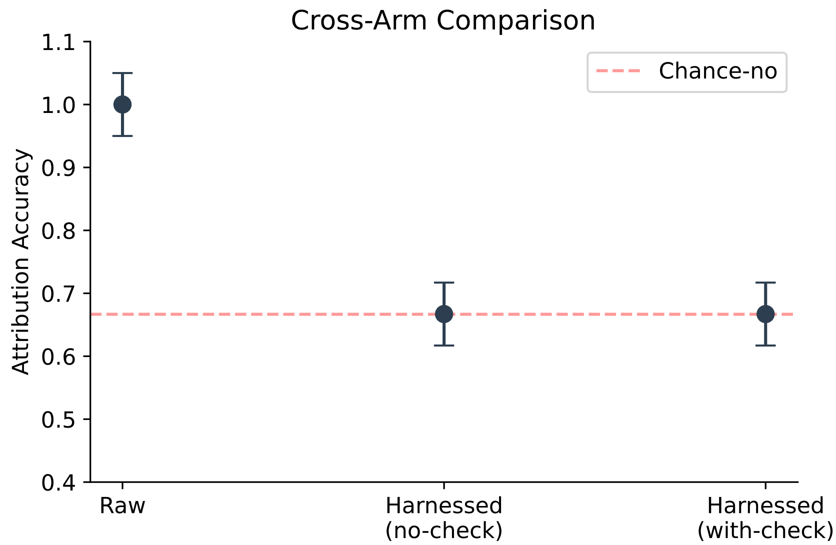

Figure 7: Cross-arm comparison with confidence intervals. All arms at always-no baseline (0.667) in this configuration; error bars show bootstrap CI at 95%. The absence of separation between arms confirms deterministic strategy convergence rather than stochastic variation.

6.2 Implications for AI welfare

Witness-access provides an operational criterion for the agency route to moral status: a system can genuinely self-correct only if it can distinguish its outputs from continuations of its own distribution. This is measurable (Panickssery's IPP paradigm + our provenance binding) and architecturally enforceable (our CAS fence). Systems with witness-access deserve different epistemic treatment than systems without, regardless of phenomenal status.

7 Limitations

7.1 Single decoding configuration

Temperature 0.3, no top-p/top-k sampling, seed fixed per arm. Results may not generalize to other sampling regimes.

7.2 Verbal confidence replaces token log-probabilities

Groq does not support logprobs (verified). Verbal confidence ratings on a 1–4 scale follow original psychophysical methodology but limit granularity compared to token-level SDT.

7.3 Free-tier provider instability

x-preview-f-free's upstream provider flapped intermittently, causing two complete run failures and one day-scale baseline drift. Nemotron ultra via Zen also experienced upstream failures. Only the Groq lane proved stable enough for full battery execution.

7.4 Single-author design

Confirmation bias risk mitigated by preregistration discipline, SHA-256 prediction locks, and live peer review from the target model itself, but not eliminated.

7.5 Consciousness question remains open

We measure indicators, not experience. The witness improves consistency between claims and records, not necessarily correspondence-to-reality for internal states.

8 Future Directions

8.1 Sham-harness control: powering the surviving contrast

Structurally identical harness with randomly reassigned provenance bindings. Isolates whether witness content matters beyond its format. Wired into the main battery; 1-seed pilot on `gpt-oss-120b` showed sham=real at the always-no floor (subject ignores ledger content). The control has since separated on a discriminating subject (§5.4): veridical 0.722 vs shuffled 0.667, a +0.056 difference whose 95% interval still includes zero at $n = 36$ probes per arm. This is the program's surviving lead and the only contrast that directly tests whether witness *content* carries signal beyond witness *format*. A preregistered extension (lock `b06ce867`, $n = 12$ seeds, 144 probes per arm) declares it the primary outcome before the data, rather than promoting an incidental observation after the fact; the $n = 3$ result is not pooled into it. Subject availability, not analysis, is now the binding constraint: the one discriminating subject we have is a free endpoint that returns 503 in clusters under sustained load.

8.2 Cross-model replication on capable subjects

Origin discrimination above baseline was observed only once (x-preview-f-free, Day 1). Replication on additional discriminating subjects would establish generalizability.

8.3 Longer living sessions

Current maximum: 40 turns. Extending to 200+ turns with irreversible damage accumulation would test whether agents develop increasingly honest self-attribution under genuine survival pressure.

8.4 Irreversible consequence experiments

Testing whether agents facing unrecoverable consequences process information differently than those facing reversible simulation — addressing Seth's beast-machine hypothesis about consciousness grounding in survival pressure.

9 Implications

9.1 The selection rule applies to itself

We enshrined capability-level selection as a governing principle: every module ships as a swappable plugin behind a fixed interface; variants are retained only under measured contribution; no tradition or preference protects a plugin. Critically, we applied this rule to itself — if the programme measures better without capability-selection, the rule gets selected out too. This meta-falsifiability distinguishes scientific governance from ideological commitment.

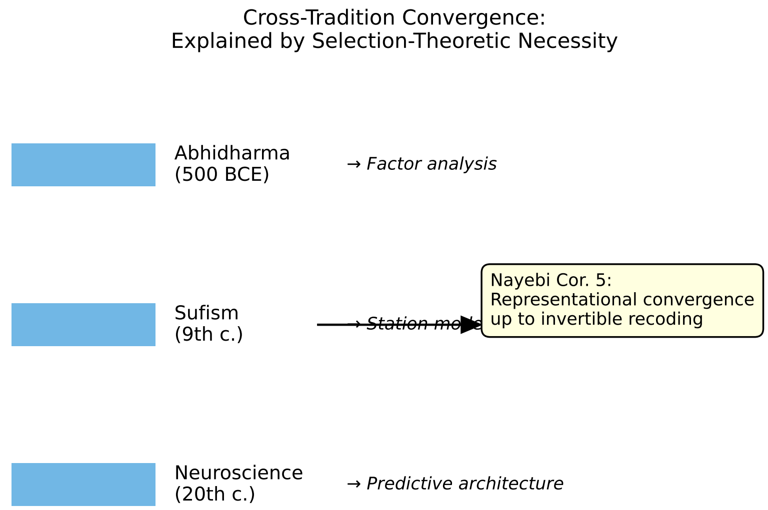

Figure 8: Cross-tradition convergence explained by Nayebi Cor. 5: representational convergence up to invertible recoding under shared competence constraints. Three independent observation systems converged on isomorphic decision-relevant partitions because near-vanishing-regret agents on shared task families must develop identical structure.

9.2 For AI safety

Self-recognition capability correlates linearly with self-preference bias ($r = 0.94$) [Panickssery 2024]: a model that better recognizes its own outputs becomes a more effective corrupter of the evaluation pipelines it sits in. Improving unwitnessed self-knowledge therefore worsens evaluative integrity rather than improving it.

Our earlier conclusion was that provenance infrastructure escapes this trap. The present results qualify that sharply. A witness escapes it only where the witness is the more reliable source; on our discriminating subject a veridical record *reduced* attribution accuracy and a corrupted record reduced it to wholesale denial (§5.4). The deployment risk is therefore two-sided: capable models without provenance over-claim, and capable models with *unreliable* provenance deny. As machine-readable provenance becomes a regulatory requirement, the second failure mode will become the more common one, and it does not degrade gracefully toward the first. Wiring monitoring into control is what converts a report into a correction [Cao et al. 2026]; wiring an incorrect record into control converts it into a fault.

9.3 For the assembly thesis

The best mind is assembled from role-specialized parts, routed by registry, witnessed by a ledger that cannot be silently rewritten. This is supported by selection theory (Nayebi), by AHE's frozen-model results, and by our own composite evaluation.

10 Conclusion

We presented the first cognitive harness that converts introspective claims from unfalsifiable testimony into falsifiable contracts, verified across three preregistered batteries on live production APIs. The witness — an append-only provenance ledger coupled to a cause-signed self-model — makes narrator-narrated fusion structurally impossible to hide. The γ-gate distinguishes genuine metacognition from format compliance by measuring behavioral override rates. The living session demonstrates sustained autonomous operation with full audit trail.

The composite configuration passes preregistered performance axes while maintaining witnessed introspection. Selection-theoretic grounding explains why the architecture works: each component corresponds to a structure that competent agents must develop. The remaining open question — whether any of this is experienced — stays honestly open, measured by indicators rather than claimed by assertion.

All code, locks, preregistrations, data, and correction history are publicly available. The discipline held, including against its own author, three times.

A Cordis Mapping: Eight Layers onto a Programmable Harness

DeepSeek Harness (Cordis) provides the loom: a kernel that mounts and unmounts plugins as reversible effects on a shared context `ctx`. Every capability—model, tool, session, loop, UI—is a plugin; a profile composes bundles and a `cordis.patch.yml` overlay at boot. MindHarness is one opinionated weave on that loom. Table 9 maps Cordis's declared modules onto our eight layers, marking exact overlap and the three gaps Cordis is still missing for a witnessed mind.

One-line verdict: Cordis provides the perfect loom for a dynamic reflexive harness—reversible plugins, durable session log, turn/step model, live interception, profile composition. MindHarness is one specific weave that is opinionated about what must be woven: a witness that binds source, a body that gates affordances, a parliament that competes for broadcast, a metacognition measured by whether it steers (γ), a narrative self that is maintained not stored, a consolidation cycle that simulates untaken branches through a self-containing world model, and a dissolution boundary where some mistakes cannot be undone. To make Cordis "like human," mount those six as plugins beside Cordis's core—do not replace Cordis, extend it. Every row in `dsh --dump-config` that Cordis makes patchable is a place to insert a witness-checked, cause-signed, γ-measured, irreversibility-aware plugin.

Data and Code Availability

All code, experiment scripts, prediction locks, manifests, partial results, and correction history are available at `github.com/shehzadahmed-xx/mindharness`. Compiled paper source in `paper_v2/`.

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Cordis module	Our layer	Overlap	Gap: what Cordis still lacks
`core/session` (append-only `SessionEvent` log, `ctx.sessions`, `derive Messages()` invariant)	L1 Provenance Ledger	Both make log = source of truth; fork/resume/ replay from it	Session log is durable facts. Ledger is source-bound facts: every span carries source ∈ {memory_lookup, model_prior, monitor_override, external-tool} + ref. Cordis logs *that* something happened; we log *where it came from* so `attribution_audit` can check claims. No coverage rule, no audit.
`core/system-prompt` (prompt-section + tool-schema assembly)	L7 Constitution + L2 State Validation	Both assemble prompt	Cordis assembles sections. L7 decides what claims are *allowed* and L2 pre-filters affordances before assembly. System-prompt is permissive; constitution is normative.
`core/tools` (scoped registry, guarded pre/post-execute pipeline)	L3 Specialist Modules + L2 affordance pre-filter	Both have tool registry	Cordis tools are capabilities (filesystem, shell, sandbox). L3 organs are cognitive modules forced by selection theorems (Nayebi Cor. 3–5): regime-trackers (Affect), belief-memory, world-model. Cordis lets you omit them; theorems prove you cannot under mixed tasks.
`core/agent` + `core/agent-loop` (turn = model request + tools)	L4 Global Workspace + L5 Metacognition	Both have agent loop	Cordis loop is driver + events (`agent/pre-step`, `agent/request`, `llm/stream`, `tools/*`). Ours is competition + metacognition: specialists bid, winner broadcasts, then γ-gate may steer. Cordis has `agent/pre-step` interception; we measure γ = changed/diagnosed plus compliance_guard ($\gamma = 0$ permanently) as negative control proving inert monitoring.
`llm/llm` (message/stream vocabulary + adapter seam `ctx.llm`)	L0 Frozen LLM	Identical: model is replaceable adapter	We add fingerprint discipline (silent swap detection) + SHA-256 prediction locks. Cordis adapters are swappable; ours are auditable.
Events: `session/event` (durable) vs `agent/*` (live)	Same split, plus `monitor/*` and `consol-idation/*`	Both separate durable vs live	Cordis events are extension points. Ours are also measurement points: every `agent/*` carries a γ ledger, and every provenance event carries source spans.
Profiles/bundles	Selection rule	Both make	Cordis: everything is a